PLO1 : Apply Java string and file handling techniques to process and manipulate data.

1 Write a program that compares two strings and checks for substring presence.

Code:

```
public class StringCompare {
    public static void main(String[] args) {
        String s1 = "Hello Java";
        String s2 = "Hello Java";
        String sub = "Java";

        if (s1.equals(s2)) {
            System.out.println("Strings are equal");
        } else {
            System.out.println("Strings are not equal");
        }

        if (s1.contains(sub)) {
            System.out.println("Substring is present");
        } else {
            System.out.println("Substring is not present");
        }
    }
}
```

Output:

Output

```
Strings are equal
Substring is present
```

2 Write a program to detect the character encoding of a given text file.

```
import java.util.Scanner;

public class CharacterEncoding {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        System.out.println("\nCharacters and ASCII values:");

        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);
            System.out.println(ch + " = " + (int) ch);
        }


        byte[] encoded = str.getBytes(java.nio.charset.StandardCharsets.UTF_8);

        System.out.println("\nEncoded values:");
        for (byte b : encoded) {
            System.out.print((b & 255) + " ");
        }


        String decoded = new String(
            encoded,
            java.nio.charset.StandardCharsets.UTF_8
        );

        System.out.println("\n\nDecoded string: " + decoded);

        sc.close();
    }
}
```

Output:

Output

```
Enter a string: abc

Characters and ASCII values:
a = 97
b = 98
c = 99

Encoded values:
97 98 99

Decoded string: abc
```

3 Write a program that takes user input for multiple strings and appends them using StringBuilder

Code:

```
import java.util.Scanner;

public class StringAppend {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        StringBuilder sb = new StringBuilder();

        System.out.println("Enter strings (type 'stop' to finish):");

        while (true) {
            String str = sc.nextLine();

            if (str.equalsIgnoreCase("stop")) {
                break;
            }

            sb.append(str).append(" ");
```

```
        }

        System.out.println("Appended string: " + sb);

        sb.append((String) null);
        System.out.println("After appending null: " + sb);

        String s = null;

        if (s == null) {
            System.out.println("String can store null.");
        }


    }
}
```

Output:

Output

```
Enter strings (type 'stop' to finish):
abhishek
umang
stop
Appended string: abhishek umang
After appending null: abhishek umang null
String can store null.
```

4 Write a program to split a paragraph into individual sentences
Code

```
import java.util.Scanner;

public class SentenceSplit {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter a paragraph:");
        String paragraph = sc.nextLine();

        String[] sentences = paragraph.split("(?<=[.!?])\\s+");

        System.out.println("\nIndividual sentences:");

        for (int i = 0; i < sentences.length; i++) {
            System.out.println(sentences[i]);
        }


    }
}
```

Output

```
Enter a paragraph:
hi. hello! how are you?

Individual sentences:
hi.
hello!
how are you?
```

4 Write a program to convert a date object to a string in a specific format
Code

```
import java.util.Date;
import java.text.SimpleDateFormat;

public class DateToString {
    public static void main(String[] args) {

        Date date = new Date();
```

```
        SimpleDateFormat sdf =
                new SimpleDateFormat("dd-MM-yyyy");

        String str = sdf.format(date);

        System.out.println("Date object: " + date);
        System.out.println("Formatted date: " + str);
    }
}
```

Output

```
Date object: Fri Oct 23 07:47:33 GMT 2026
Formatted date: 23-10-2026
```

5 Write a program to insert a substring into a string at a specific position using StringBuilder

Code

```
import java.util.Scanner;

public class StringInsert {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        System.out.print("Enter substring to insert: ");
        String sub = sc.nextLine();

        System.out.print("Enter position: ");
        int pos = sc.nextInt();

        StringBuilder sb = new StringBuilder(str);

        if (pos >= 0 && pos <= sb.length()) {
```

```
            sb.insert(pos, sub);
            System.out.println("Result: " + sb);
        } else {
            System.out.println("Invalid position!");
        }

        sc.close();
    }
}
```

Output

```
Enter a string: hello abhishek
Enter substring to insert: 5
Enter position: 3
Result: hel5lo abhishek
```

6 Write a program to remove null values from an array of strings
Code;

```
import java.util.Arrays;

public class RemoveNull {
    public static void main(String[] args) {

        String[] arr = {"Java", null, "Python", null, "C++", "Hello"};

        System.out.println("Original array:");
        System.out.println(Arrays.toString(arr));

        String[] result = Arrays.stream(arr)
                                .filter(s -> s != null)
                                .toArray(String[]::new);

        System.out.println("Array after removing null values:");
        System.out.println(Arrays.toString(result));
```

```
    }
}
```

Output

```
Original array:
[Java, null, Python, null, C++, Hello]
Array after removing null values:
[Java, Python, C++, Hello]
```

PLO2 :Apply Java classes and APIs for pattern matching, character processing, and string manipulation.

1. Write a program to find all occurrences of a pattern in a string using Pattern and Matcher

Code

```
import java.util.Scanner;
import java.util.regex.Pattern;
import java.util.regex.Matcher;

public class PatternMatch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        System.out.print("Enter a pattern to search: ");
        String pattern = sc.nextLine();

        Pattern p = Pattern.compile(
                Pattern.quote(pattern));

        Matcher m = p.matcher(str);

        int count = 0;
```

```
        while (m.find()) {
            System.out.println(
                    "Pattern found at index: " + m.start());
            count++;
        }

        if (count == 0) {
            System.out.println("Pattern not found.");
        } else {
            System.out.println("Total occurrences: " + count);
        }

        sc.close();
    }
}
```

Output

```
Enter a string: hello
Enter a pattern to search: e
Pattern found at index: 1
Total occurrences: 1
```

2 Write a program to count the number of vowels in a string using the Character class.

Code

```
import java.util.Scanner;

public class VowelCount {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        int count = 0;

        for (int i = 0; i < str.length(); i++) {
```

```
            char ch = Character.toLowerCase(str.charAt(i));

            if (Character.isLetter(ch) &&
                (ch == 'a' || ch == 'e' || ch == 'i' ||
                 ch == 'o' || ch == 'u')) {
                count++;
            }
        }

        System.out.println("Number of vowels: " + count);

        sc.close();
    }
}
```

Output

```
Enter a string: abhishek
Number of vowels: 3
```

3 Write a program to check whether a given string is a palindrome or not using StringBuffer, String

Code;

```
import java.util.Scanner;

public class VowelCount {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        int count = 0;

        for (int i = 0; i < str.length(); i++) {
            char ch = Character.toLowerCase(str.charAt(i));
```

```
        if (Character.isLetter(ch) &&
            (ch == 'a' || ch == 'e' || ch == 'i' ||
             ch == 'o' || ch == 'u')) {
            count++;
        }
    }

    System.out.println("Number of vowels: " + count);

    sc.close();
  }
}
```

Output

```
Enter a string: abhishek
Number of vowels: 3
```